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Specialization:

Machine Learning

Business Focus:

HR Analytics & Labour Markets

Skill Outcome:

- Regression Modelling
- NLP & Skills Extraction
- Model Evaluation & Salary Insights

Tech Job Market Salary Prediction — ML Regression Case Study

A FSDS CASE STUDY

Project Learning Opportunities:

This project equips students with end-to-end ML skills: loading and cleaning a real global jobs dataset, engineering salary and skills features, training and tuning regression and classification models, and translating outputs into actionable talent market insights for HR and recruitment teams.

Tools and Technology Used:

• Python 3 • Scikit-learn • Pandas • NumPy • Matplotlib • Seaborn • Jupyter Notebook

Company Background

TalentPulse Analytics is a global HR-tech startup founded in 2020. The company provides data-driven workforce intelligence to recruitment agencies, enterprise HR teams, and job platforms across the USA, UK, Canada, India, and Australia. Its platform ingests live job postings, extracts structured signals from unstructured descriptions, and surfaces salary benchmarks and skills demand reports to help organisations hire faster and pay competitively.

The company's dataset covers 4,653 live job postings scraped from five countries, spanning roles including Data Engineer, Data Scientist, Machine Learning Engineer, Business Analyst, and Data Analyst across Junior, Mid, Senior, and Lead levels. TalentPulse operates a lean team of 60 data scientists, engineers, and product managers who continuously enrich and retrain the platform's salary models.

As the global tech job market has grown increasingly competitive, clients demand more accurate and explainable salary benchmarks. The company's legacy lookup-table salary model has shown growing errors, particularly for senior and remote roles. Leadership has commissioned a machine learning initiative to replace the legacy model with a robust, data-driven regression pipeline trained on the TalentPulse Jobs Dataset — 4,653 job postings spanning five countries and multiple specialisations.

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The Business Problem

TalentPulse Analytics’ legacy lookup-table salary model produces a Mean Absolute Error (MAE) of $18,500 per role — far exceeding the industry benchmark of $8,000. As global tech salaries increasingly diverge between senior and junior roles, and between remote and on-site positions, the model systematically under-estimates senior salaries and over-estimates junior compensation, eroding client trust and candidate offer-acceptance rates.

The Problem

The root cause is threefold. First, the model ignores seniority effects: Senior and Lead roles command a 45-80% salary premium over Mid-level roles that a flat lookup table cannot capture. Second, remote availability is a non-linear predictor — its effect on salary accelerates sharply for roles requiring 5+ years of experience. Third, in-demand skills such as Python, Spark, and AWS command a scarcity premium not reflected in job-title-based benchmarks. Together these gaps result in mispriced offers, lost top-tier candidates, and competitive disadvantage for TalentPulse clients.

Your Role

You are a Machine Learning Engineer at TalentPulse Analytics. Your mandate is to build a supervised regression pipeline on the TalentPulse Jobs Dataset (4,653 job postings across 5 countries) to predict average salary. You will perform full EDA, clean and engineer features, compare at least three regression algorithms, tune the best model via cross-validation, and present findings with RMSE, MAE, and R^2 metrics — plus an actionable salary benchmarking recommendation for the HR team.

TalentPulse ANALYTICS

Workforce Intelligence. Smarter Hiring. Better Outcomes.

Rationale for the Project

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Business Impact

Replacing the legacy salary lookup model with a trained regression model is projected to reduce MAE by 57%, saving TalentPulse an estimated $1.6M annually in lost client contracts due to inaccurate benchmarks.

Market Segmentation

Feature engineering on job level, remote status, skills count, and country surfaces market segments commanding 45-80% salary premiums — creating targeted compensation intelligence for recruitment clients.

Regulatory Compliance

A fully reproducible, version-controlled ML pipeline meets the audit requirements of TalentPulse's enterprise clients and reduces pay-equity compliance risk from opaque, non-data-driven compensation rules.

Project Workflow

Step 1 — Data Loading & EDA: Load Jobs CSV (4,653 rows), inspect dtypes, identify missing salary values, visualise distributions of salary_avg, job_level, and is_remote, and plot grouped bar charts by country and job_title.

Step 2 — Data Cleaning: Handle 2,358 missing salary_avg values. Remove duplicates. One-hot encode job_level, country, and is_remote. Parse and flatten the skills column.

Step 3 — Feature Engineering: Create salary_range, is_senior flag, experience_tier buckets, and top-skill binary flags (python, sql, spark, aws). Log-transform skewed features.

Step 4 — Model Training & Comparison: Train Linear Regression, Random Forest, and Gradient Boosting on 80/20 split. Tune hyperparameters via 5-fold CV GridSearchCV.

Step 5 — Evaluation: Compare RMSE, MAE, R^2 across all models. Plot feature importances. Visualise residuals by country and job level. Deliver a benchmarking memo.

Data Dictionary

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TALENTPULSE JOBS DATASET (Source: Global Job Board Scrape, 2024)

- job_id: Unique identifier assigned to each job posting.
- job_title: The title or designation of the advertised job role.
- company: Name of the organization offering the job.
- location: City, state, or region where the job is located.
- salary_min: Minimum salary offered for the position.
- salary_max: Maximum salary offered for the position.
- description: Detailed job description including responsibilities, requirements, and qualifications.
- country: Country where the job opportunity is based.
- search_keyword: Keyword used to search for or categorize the job posting.
- experience_required: Number of years of experience required for the role.
- degree_required: Educational qualification required for the position, such as Bachelor's, Master's, or PhD.
- skills: Technical and/or soft skills required for the role.
- num_skills: Total number of skills listed in the job posting.
- job_level: Seniority level of the position, such as Entry-Level, Mid-Level, or Senior-Level.
- is_remote: Indicates whether the job can be performed remotely (True) or not (False).
- salary_avg: Average salary for the position, calculated as the mean of salary_min and salary_max.
- has_salary: Indicates whether salary information is available in the job posting (True) or missing (False).

Key Analytical Questions

Exploratory Data Analysis Questions

- What is the distribution of salary_avg and does it require log-transformation before modelling?
- Which features show the highest Pearson correlation with salary_avg?
- How does salary_avg vary across job_level and country categories? Plot grouped boxplots.

Data Preparation & Feature Engineering

- How should the 2,358 missing salary_avg values be handled? Should you impute, drop, or model them separately? Justify your strategy.
- Which engineered features (salary_range, is_senior, num_skills, top-skill flags) improve model R^2?
- Which job_level categories should be collapsed or binarised given their frequency distribution?
- Does a log-transform on salary_avg improve residual normality? Show before/after Q-Q plots.
- Which features have VIF > 10, indicating multicollinearity that should be removed?
- How does the train/test score gap change as Random Forest max_depth increases from 3 to 20? At what depth does overfitting begin?
- What regularisation strength (alpha) minimises Ridge Regression CV error on this dataset?
- Segment roles into "Standard" (Mid/Junior) vs "Premium" (Senior/Lead) — how does model accuracy differ between segments?

Business & Model Evaluation Questions

- Which country or job_level shows the highest systematic salary underestimation by the model? Quantify the gap.
- Which algorithm achieves the lowest RMSE on the test set: Linear Regression, Random Forest, or Gradient Boosting?
- Are prediction residuals clustered by country or seniority level? Plot them to identify systematic bias zones.

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Let's Dive in

Please note

When you are done with this case study:

Kindly **share your work on WhatsApp**, and don't forget to **add it to your portfolio** or **share on LinkedIn**

Remember to tag @10Alytics and use these #MachineLearning #HRAnalytics #DataScienceWith10Alytics